

Assignment-01-AT-01

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ITA0614 - Machine Learning

1) In a real world medical image classification task for detecting breast cancer, a deep learning model integrated with the k-nearest Neighbour algorithm is applied for binary classification (malignant vs benign). The Dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positive (TP), 190 True Negative (TN), and 30 False Positive (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and specificity of the classifier, explaining its effectiveness in medical Diagnosis.

Sol:- Given,

$$TP = 210 \quad FP = 30$$

$$TN = 190 \quad FN = ?$$

$$\begin{aligned} \text{Total Dataset} &= 1800 \text{ (malignant)} + 1800 \text{ (benign)} \\ &= 3600 \text{ samples} \end{aligned}$$

$$\text{Test Set (20\%)} = \frac{20}{100} \times 3600 \Rightarrow 720 \text{ samples.}$$

False Negatives (FN)

$$FN = \text{Total Test} - (TP + TN + FP)$$

$$FN = 720 - (210 + 190 + 30)$$

$$= 720 - 430 = 290$$

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{T_{\text{Total}}}$$

$$\text{Accuracy} = \frac{210 + 190}{720} = \frac{400}{720} = 0.556 (55.6\%)$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Precision} = \frac{210}{210 + 30} = \frac{210}{240}$$

$$\text{Precision} = 0.875 (87.5\%)$$

Sensitivity (Recall)

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{210}{210 + 290} = \frac{210}{500}$$

$$\text{Recall} = 0.42 (42\%)$$

Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Specificity} = \frac{190}{190 + 30} = \frac{190}{220} = 0.864 (86.4\%)$$

In medical applications, high recall is more important than precision, because missing a cancer case is far more serious than false alarm.

2)

Examine the performance of a machine-learning model. using its training and validation loss values over the multiple epochs.

Epoch	Training loss	validation loss
1.	2.8	2.4
2.	2.5	2.2
3.	2.2	2.0
4.	2.0	2.1
5.	1.8	2.3
6.	1.6	2.6
7.	1.5	2.9
8.	1.3	3.2

(i) Identify the epoch at which the model begins to overfit.

upto Epoch 3:

* Training ↓ and validation ↓ → good learning.

From Epoch 4 onwards:

* Training loss ↓ (model improving on the training Data).

* validation loss ↑ (performance weakening on unseen data).

⇒ overfitting starts at epoch 4.

(ii) Suggest two effective techniques to reduce overfitting in this model.

Early stopping

* stop training when validation loss starts increasing.

* In this case → stop at epoch 3.

Regularization:

- * Add constraints to prevent model from the memorizing data.

- * Examples.

- ⇒ L2 Regularization (Ridge)

- ⇒ Drop out (in Deep learning)

(iii) Discuss how overfitting affects the models' Generalization ability on unseen data.

- * The model memorized training data instead of learning patterns.

- * Performance: Very well on Training Data and Poorly on new/unseen Data.

- * In this case training loss keeps Decreasing (1.3 at epoch 8) and Validation loss increases (3.2 at epoch 8) and validation loss increases.

- * This means model is not Generalizing predictions on real world Data will be inaccurate.